

3) Induction motor:- 1) The IM which converts electrical energy into mechanical energy is called Induction motor.

2) I.M can operate into two modes $\rightarrow$ Motoring mode $N < N_s$
 $\rightarrow$ Generating mode $N > N_s$
 N_s = synchronous speed.

3) IM is called asynchronous speed. At ' N_s ' it can ^{converts} ~~generate~~ electrical power or mech power ^{operates/converts}.

4) By synchronous m/c can ^{only} produce electrical power mech power at ' N_s ' system.

I.C.M

T/F
 $\rightarrow$ 1) Stator winding (primary)

2) A generalized transformer with short circuit rotating secondary.

3) T/F does not change freq.

4) Magnetic circuit is continuous.
magnetic circuit (a No air gap)

5) Single exciting m/c.

6) T/F operates mutual induction principle.

7) I.M is single exciting m/c just like T/F.

8) I.M operates mutual induction principle.

1) The rotor winding (secondary winding) It is always rotating and always short circuit. short circuit rotating secondary

3) Here (I.M) power transform from stator to rotor at different freq.

4) frequency changing m/c.

5) Magnetic circuit is composite magnetic circuit (having air gap).

6) I.M has discontinuous or composite magnetic circuit.

Synchronous generator

1) Generate electrical power at ' N_s ' speed means ' $N = N_s$ '
 Const speed = synchronous speed.

2) Generally used const speed prime mover.
 (Steam turbine, diesel engine)

Large hydro turbine by using governor arrangement we can get const speed prime mover.

Induction generator

1) Speed $N > N_s$

2) In order to produce synchronous speed variable speed prime mover is used ex:- windmill, or fluctuating prime mover.

3) Low head hydropower station less than 30 meter head has variable speed prime mover.

$\rightarrow$ Induction motor has shunt type characteristic.

$\rightarrow$ Induction motor can be replaced by d.c shunt motor as same characteristic.

Production of rotating magnetic field in 3 ϕ system:

When even 3 ϕ winding are physically displaced by $2\pi/3$ (120°) electrical and being fed from 3 ϕ current which are also displaced by $2\pi/3$ electrical rad with respect to time angle ωt then the resultant mmf with const amplitude whose value = 1.5 times max value of mmf to any phase and it rotating a synchronous speed will be produced in the air gap.

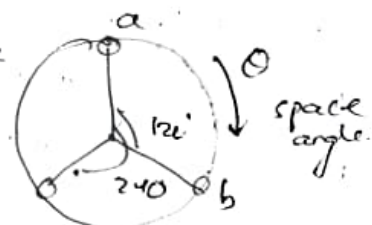
→ Condition should be satisfy: - For rotating field in 3 ϕ system

1) There is physically displacement of 120° electrical between 3 ϕ winding. 2) There must be time displacement of 120° electrical between the 3 ϕ current flowing through the winding.

Why Rotating magnetic is not available in T.F? Phase
As the T.F winding is not displaced physical 120° displacement.

→ 3 phase winding displaced 120° space angle

→ 3 ϕ supply with time angle varying
 $\omega t = 120^\circ$



physical location
 $a_{\text{phase}} = N_{ph} \cos \theta$ simplification say may be taken $\sin \theta$

$$b_{\text{phase}} = N_{ph} \cos (\theta - 120^\circ)$$

$$c_{\text{phase}} = N_{ph} \cos (\theta - 240^\circ)$$

$$i_a = I_m \cos \omega t \quad i_b = I_m \cos (\omega t - 120^\circ) \quad i_c = I_m \cos (\omega t - 240^\circ)$$

$$F_a = N_{ph} \cos \theta \cdot I_m \cos \omega t$$

$$= F_m \cos \theta \cdot \cos \omega t$$

$$F_b = N_{ph} \cos (\theta - 120^\circ) I_m \cos (\omega t - 120^\circ)$$

$$= N_{ph} I_m \cos (\theta - 120^\circ) \cos (\omega t - 120^\circ)$$

$$F_c = N_{ph} \cos (\theta - 240^\circ) I_m \cos (\omega t - 240^\circ)$$

$$= N_{ph} I_m \cos (\theta - 240^\circ) \cos (\omega t - 240^\circ)$$

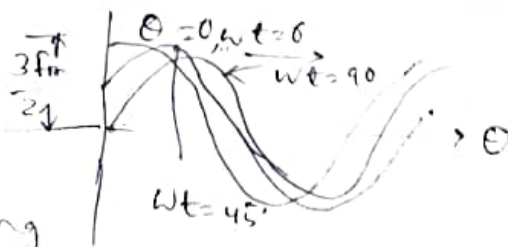
$$F_R = F_a + F_b + F_c$$

Resultant mmf

$$F_R = \frac{3}{2} F_m \cos (\theta - \omega t)$$

The resultant mmf is rotating with respect to space.

So it is called rotating



$$J_0 = J_0 e^{-J_0/2} \eta e^{-J_0}$$

on 5/10/10

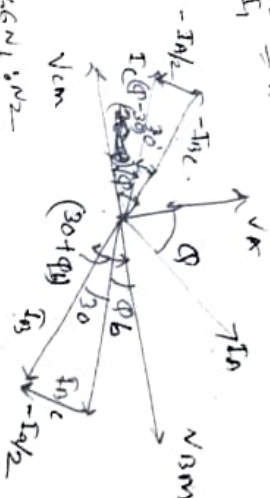
$$\frac{1.15}{1.15} = 1.15$$
$$T_c = +50^\circ\text{C} + 10^\circ\text{C} \cdot \log \frac{1}{\text{g/sec}}$$

To check back to

$$i_a = 1 \text{ A}$$
$$I_A = 1.015 A$$
$$T_{AC} = i\alpha$$
$$T_c = \sqrt{\frac{2}{\pi}} \frac{1}{\omega_p}$$

then least of main for $(30^\circ - \phi)$

→ of the load put on one half of the load but one operator at a time as the load and another half of the load

$$\cos(30^\circ + \theta) \text{ and } \sin(30^\circ + \theta)$$


2 sets from factor of
 → utilize connection with
 Scott connection to it.
 two identical to it.
 26.6% → of two
 dissimilar → one with
 other with.

$$-2N^2 + 6N - 6$$

different from
N1: N2 another with
for Scott connection
Q2.87.

and correction factor is 0.201
the utilization factor = $\frac{(K_{uh}) \text{scot}}{\text{available kva}} = \frac{\sqrt{3} V_L I_L}{2 V_L I_L} = 0.866$

→ 12127281000
→ 12127281000 open 22

$$\frac{V_{eff} = \frac{V_{eff}}{\sqrt{3}}}{1.84}$$
$$\frac{0.4}{0.866} = 0.4618$$

Parallel operation of two or more small gas engines

1) The parallel operation of power supply.

the reliable of person 11) then of the system.

2) Parallel operation

3) Cost of standard
condition:

~~Necessary~~ ^{conceded} ~~must be properly~~ ^{conceded with regard to}

pelant

mmf or travelling mmf: it is not stationary
→ 'ω' determine the space of resultant mmf wave.

Remember

for P poles of machine, there $P/2$ pairs of poles
magnetic field rotate through $\frac{f}{P/2}$ or $\frac{2f}{P}$ rotates per second.

$$f = \text{cycles of current} \\ N_s (\text{synchronous speed}) = \frac{f}{P/2} = \frac{2f}{P} \text{ rps} = \frac{120f}{P} \text{ rpm}$$

ω = angular freq.

$\omega_e = \left(\frac{1}{2}\right) \omega_m$ angular velocity.
 $\rightarrow$ mechanical rad/sec
 $\rightarrow 2\pi f = \frac{1}{2} \times \frac{2\pi N}{60}$
 $\rightarrow$ electrical (rad/sec) $\rightarrow$ $N \text{ in rpm} = \frac{120f}{p}$

Generalized condition that must be satisfy, we get rotating magnetic field by using any no of phases > 2 .

1) m phase winding where ($m > 2$) must be physically displaced by $\frac{2\pi}{m}$ rad electrical in the space. 2) Condition: The time displacement between m phase current flowing through the m winding must be equal to $\frac{2\pi}{m}$ rad electrical.

2 ϕ system
 $m = 2$

$\frac{\pi}{2}$ rad $\rightarrow$ θ winding
 $\frac{\pi}{2}$ " $\rightarrow$ ωt current

$F_R = F_m \cos(\omega t - \theta)$
 $N_s = 120f/p$

$F_a = N_p h \cos \theta I_m \cos \omega t = N_p h I_m \cos \theta \cos \omega t$

$F_b = N_p \cos(\theta - 90^\circ) I_m \cos(\omega t - 90^\circ) = N_p I_m \cos(\theta - 90^\circ) \cos(\omega t - 90^\circ)$
 $= N_p I_m \sin \theta \sin \omega t$

$\Rightarrow F_e = F_a + F_b$

$\cos A \cdot \cos B = \frac{1}{2} [\cos(A+B) + \cos(A-B)]$

Double field Revolving theory:

$F_a = F_m \cos \theta \cos \omega t = \frac{F_m}{2} \cos(\theta - \omega t) + \frac{F_m}{2} \cos(\theta + \omega t)$

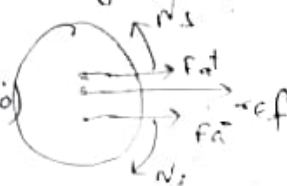
Phase a
 Rotating N_s $\rightarrow$ forward rotating mmf due to F_a^+ + F_a^- (Backward rotating mmf.)
 (but opposite direction)
 Rotating N_s

Phase b

$F_b = F_m \cos(\theta - 120^\circ) \cos(\omega t - 120^\circ)$

$= \frac{F_m}{2} \cos(\theta - \omega t) + \frac{F_m}{2} \cos(\theta + \omega t - 240^\circ)$

$= F_b^+ + F_b^-$
 $(N_s) (N_s)$



Phase c

$F_c = F_m \cos(\theta - 240^\circ) \cos(\omega t - 240^\circ)$

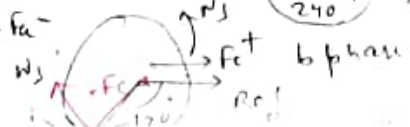
$= \frac{F_m}{2} \cos(\theta - \omega t) + \frac{F_m}{2} \cos(\theta + \omega t - 120^\circ)$

$= F_c^+ + F_c^-$



Resultant F_e 1 ϕ

back ward mmf = zero



ϕ_f ϕ_b

$T_f \rightarrow \leftarrow T_b$

Net $T=0$ (No starting torque)

but ϕ

ϕ_f $\phi_b \rightarrow$ absent

$\downarrow$
 T_f $T_b \rightarrow$ absent

Net $T=T_f$

so self starting

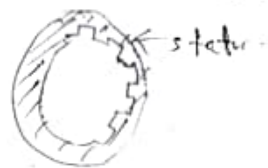
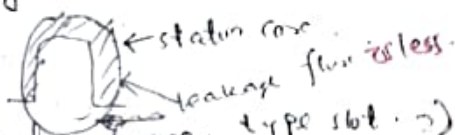
According to double field revolving theory, rotational mmf due to 3 ϕ phases can be divided into two opposite rotating mmf both rotating opposite direction. But in synchronous speed. All the 3 forward rotating mmf due to 3 phase or 2 space phase connected that gives rise a resultant forward rotating mmf in the airgap. The backward rotating mmf due to 3 ϕ are displaced by 120° apart. Therefore they are added to zero. That means a 3 ϕ system only forward rotating mmf is present. And Backward mmf is completely absent. So only forward flux is present. Backward flux is absent.

Construction Detail: 1) Stator Stationary part. 2) Rotor Rotating part.

Stator: - Slotted stator is used.
CR is used in transformer.

slot: 1) Open type 2) Semi open
2) completely close type slot.

Open type:



1) former wound coil or former type slot. 2) Easy access.

3) Low leakage reactance.

Draw back: 1) The average airgap flux is more.
So excitation current required is more.

$$X_{\text{ind}} = \frac{\Phi}{I_f} = \frac{N_f \mu_0 \mu_r}{R_f}$$

where Φ is the average airgap flux, N_f is the number of field winding turns, μ_r is the relative permeability of the field winding, and R_f is the reluctance of the field winding.

2) Non uniform reluctance. So flux

distribution is non uniform. So the harmonic flux is present. So vibrations more noise, operation is not smooth.

3) Wide

Yoke: I.M $\rightarrow$ used mmf (less)
[Used as protection cover]
[Magnetic frame / Yoke]

Narrow open type: 1) Used small

former wound coil whose dia is

less than 'd'. 2) Average air gap
is reduced. So I_e required is less.



3) The flux is more or less uniform. Then harmonic is less or
more. operation smooth and silent. Vibrations reduced. Noise
reduced. $\rightarrow$ leakage flux is strong. So leakage reactance more

Closed type: 1) More labour required as

than to turn winding is done

2) The average airgap is maximum. (I_e is less)



3) Flux distribution is uniform, as air gap is uniform.

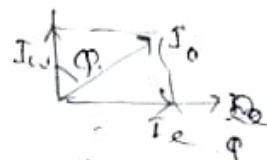
4) Harmonic torque is less, less noise

open semi-closed closed.

1) $I_e > I_e > I_e$ (magnetising / exciting current)

as an gap is more open, the semi-closed then closed.

2) $\cos \phi_0 < \cos \phi_0 < \cos \phi_0$
(No-load p.f.) less (poor)



3) $X_L < X_L < X_L$

(leakage flux is less) as a gap more



leakage flux.

$$L = \frac{N \Phi}{I} \quad X_L = 2\pi f L$$

4) Harmonic > harmonic > harmonic
(Noise) > Noise > Noise

Flux is uniformly distributed

→ As open type slot required more magnetising component of current and also the operating is not smooth and salient. The open type slot are not preferred for IM. Only semi-open or completely closed type slot are preferred.

→ Open type slots are generally preferred in synchronous m/c and also in d/c m/c.

→ In practical semi-closed slots are preferred.

$$\Delta X_s = X_a + X_{sl}$$

→ Steady state stability $\propto \frac{1}{X_s}$
→ Rippling factor $\propto \frac{1}{X_s}$

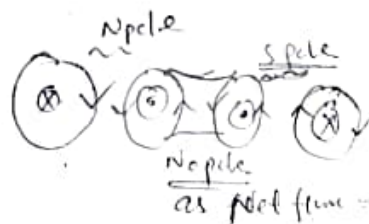
Short-circuit current is increased $\propto \frac{1}{X_s}$

$$\text{SCR} \propto \frac{1}{X_s}$$

→ Commutation is increased in d/c m/c. So commutation delay occurs due to inductance. Cause more sparking. So to improved commutation open slot are used in armature of d/c m/c. Armature reaction is decreased.

No of poles determination:

of two consecutive conductors or coils carrying current in opposite directions. Then there will be a pole formation between them.



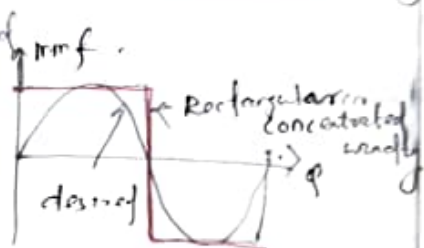
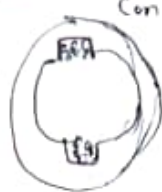
as Net flux = zero

Other hand:- Two consecutive conductors or coils carrying current in same direction. Then there will not be pole formation between them.

Station: Let 180° condenser, distributed 3P, and 4P No of poles

$$\frac{180}{3 \times 4} = 15 \text{ roof conductors/phases/pole}$$

37 15 conductors are placed in a single slot is called concentrated winding
 38 15 " " " different " " distributed winding



→ The shape of air gap mmf produced, with concentrated winding method is rectangular, which is very much deviated from desired sinusoidal wave.

✓ The shape of air gap mmf produced with distributed winding ~~per pole~~ is at stepped view with an approximate as trapezoidal view. and this is very much near to desired sinusoidal space mmf wave.



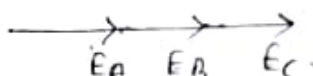
→ Concentrated tendency diversified diversity



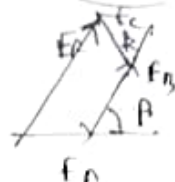
→ The emf induced
a, b and c
are in phase.



→ phase difference



The resultant emf is arithmetic
sum.



$$F_R = \vec{F}_A + \vec{F}_B + \vec{F}_C \quad (\text{Vektorsum})$$

⇒ Distributed factor.

$$\text{spread factor (kd)} = \frac{\text{emf induced in distributed winding}}{\text{emf induced in concentrated winding}}$$

If winding are concentrated evenly so emf = 4.44 gf/Hz.

Emf induced in distributed winding = $4.44 N_{ph} \Phi_m f \times k_f$

$$K_d = \frac{S_{2n} m p / 2}{m S_{2n} p / 2}$$

$$m = \text{No of slots / pole pair}.$$

→ $P = \frac{\text{slots}}{180}$ (electrical degree)
 $\frac{\text{slot/pole}}$

m β = phase spread angle

if β is very small.

$$\Rightarrow \sin \alpha / 2 = \alpha / 2$$

12. Group 1

$$K_d = \frac{\sin m\beta/2}{m\beta/2} = \frac{\sin d/2}{d/2}$$

$$K_{d_n} = \frac{\sin mn\beta/2}{m \sin \beta/2}$$

$$K_d > K_{d5} > K_{d7}$$

fundamental

g.f. $m\beta = 120^\circ$

$$K_{d3} = \frac{\sin \frac{3 \times 120}{2}}{3 \sin \frac{120}{2}} = 0$$

3rd harmonic is absent

As the order of the harmonic increase the value distribution factor

is decreases.

$$E_{f/\text{phase}} = 4.44 N_{ph} \Phi_m f \times k_{df}$$

$$E_s (\text{concentrated winding}) = 4.44 N_{ph} \Phi_m f (1)$$

$$E_s (\text{distributed "}) = 4.44 N_{ph} \Phi_m f \times k_{df}$$

As the order of harmonics increase the value of distribution factor is reduced.

$$\text{where } k_{df} < 1$$

By distributing the winding in different slot along the periphery the harmonic content in the emf can be reduced.

1) Shape of mmf air gap is sinusoidal and harmonic content is reduced in the induced emf in distributed winding.

So I.M and synch m/c winding is always distributed winding.

→ full pitch winding ✓ Short pitch or fractional pitch winding -
Cord has two coil sides

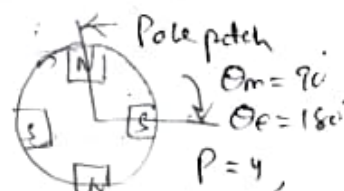


$$180^\circ = \text{coil span} = \text{one pole pitch.}$$

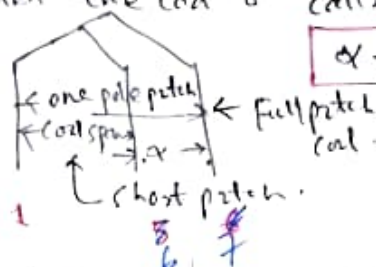
Pole pitch = Distance between two consecutive pole in electrical degree in one pole pitch.

One pole pitch = 180° electrical.

$$\theta_e = \frac{P}{2} \theta_m$$



The distance between two coil sides of a coil is equal to one pole pitch = 180° electrical then the coil is called full pitch coil.



$\alpha = 180^\circ - \text{coil span}$ (electrical) of the short pitch winding.

$$\text{coil span} = \frac{5}{6} \times 180^\circ = 150^\circ (\text{short pitch coil}).$$

$$\alpha = 180^\circ - 150^\circ = 30^\circ$$

→ If the distance between two coil sides of a coil is intentionally reduced distance less than 180° electrical position. Then this type of coil is short pitch coil.

The emf induced in full pitch coil are arithmetic sum of two coil. So the resultant emf = $E_s + E_s = 2E_s$ (arithmetic sum).

Then resultant emf induced in short pitch winding.

$$E_R = 2E_s \cos \frac{\alpha}{2}$$

Chordang. factor, pitch factor (k_p) = $\frac{\text{emf induced in each pole coil}}{\text{emf induced in each pole coil}}$

$$k_p = \frac{\text{Vector sum}}{\text{Arithmetical sum}} \Rightarrow k_p < 1$$

$$K_p = \frac{2 E_s \cos \alpha/2}{2 E_s} = \boxed{\cos \alpha/2}$$

Winding factor = $\frac{2E_s}{K_w} = K_p K_d$.

$$1/\rho h = 0.044 \text{ Npu } \rho m f \text{ Kw}$$

$$k_{p_n} = \cos \alpha / 2 \quad k_{p_f} > k_{p_5} > k_{p_7}$$

$K_{p1} = \cos \alpha / 2$. $K_{pf} > K_{p5} > K_{p7}$.
 ✓ The order of harmonics increases the value of pitch-factor reduces
 → By employing short-pitch winding the harmonic content in the
 emf can be reduced.
7th harmonic is completely eliminated

$$\frac{n\alpha}{2} = k \frac{\pi}{2} \Rightarrow \alpha = \frac{\pi}{n}$$

→ To eliminate 7th term $\alpha = \pi/7$

→ To eliminate 7th harmonic $\alpha = \pi/7$
→ For given α value only eliminated one harmonic.
→ For given α value, such that $\alpha \neq \pi/7$

$\Rightarrow$ By properly choosing α values such that $\frac{m}{2} \geq \frac{k}{2}$, where $k =$ odd integer. Any one desired n th harmonics can be completely eliminated from the system.

✓ eliminated from the system.
short patch. coil required less "overhang" or end connection, so saving of Cu. But in practical to produce same amount of emf the no of turns is increased. So the increased Cu is nullify by the less overhang Cu. So the copper is saved.
statically short patch and distributed.

by the less over hand in so copper is short.
 → Windy. are practically short pitch and distributed.
 → By employing short pitch winding reduce the amount of ~~to~~ cc
 required for the winding redices as compared to full pitch
 winding.

Construction Details of Motor:

- 1) Squirrel cage motor.
- 2) Wound rotor / slip ring T.M. $\rightarrow$ Both have same stator construction.

Cage Rotor: Endring made in high conductive forged copper.

high conductive forged copper
→ rotor is having more thickened laminations
compared to stator → very loss.
→ As the f of rotor is very low.
 $f = 50$ (2 to 3 Hz) → so the losses are very low.

Feature:- The same no. of stator poles automatically induced
in the rotor poles i.e. no. of stator poles = no. of rotor poles



in the sq. case rotor by means of induction. 2) The square cage
rotor can respond automatically to changes ^{equal no} in stator of
poles.

As rotor are faces smooth then air gap is even between stator and rotor therefore the exciting current required is less and the m/c has more no load p.f. Full load p.f. is definitely high.

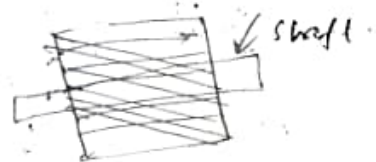
$$\uparrow T_{max} \propto \frac{1}{X_2} \downarrow \text{ (leakage reactance) }$$

→ As winding overhead are absent. The leakage flux and hence leakage reactance of the rotor is less

$$\downarrow T_{st} \propto R_2 \downarrow \text{ (All bars are connected in parallel so } R_2 \text{ is very less) }$$

→ As sq. cage rotor has low rotor resistance, T_{st} is poor and its starting current is high. But excellent running ch. such as more torque, less magnetizing component current, good no load and full load p.f. But this type of m/c has poor starting ch. such as high starting current and low T_{st}

→ The main purpose of skew rotor slot is to make the air gap flux disturbance more or less uniform thereby reduces harmonic torque in I.M.



→ The harmonic torque cause → cogging and magnetic locking.
 1) By skewing the rotor slot cogging or magnetic locking phenomena can be reduce.
 2) Chattering phenomena can be reduced.
 3) It also has to give smooth and satisfactory operation of the I.M.

Wound Rotor:-

The condition must be satisfy while winding wound rotor.
 1) Stator 3φ 1) If though the stator is wound for 3φ rotor need not wound for 3φ, But it can be wound for any no of phases except single phase. [But in practice the rotor is also wound for 3φ only].

If stator pole ≠ rotor pole ⇒ no torque is produced in I.M.
 But in sq. cage I.M. automatically change take place the no of pole.

2) The rotor must be wound for same no of poles as the stator winding. This type of rotor can't response automatically to changes in stator no of poles

→ The rotor 3φ winding must be connected in Y (star) irrespective of whether it stator winding connected in star or delta.

→ Function of external resistance 1) It increase the

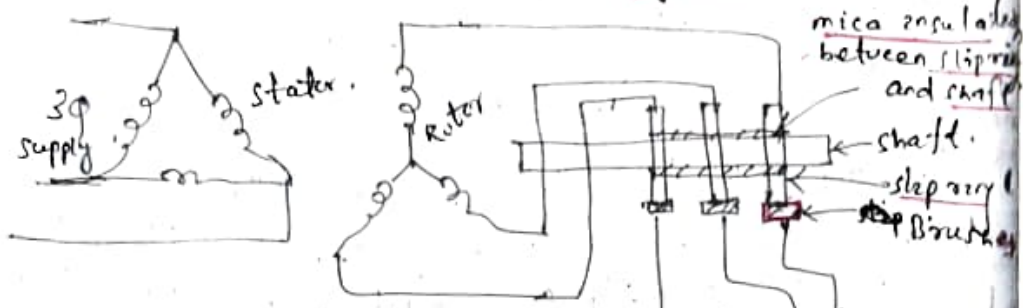
starting torque produce by the m/c.

2) It also limits the starting current.

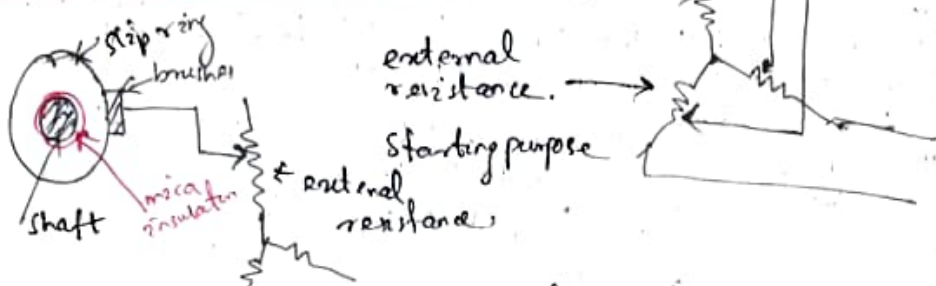
Therefore no external starting methods are required.

- 3) $P \cdot f = 1 \cos \phi = \frac{R}{Z}$. So it increases the starting p.f. of motor. ($\frac{R}{Z}$ is very low in sq. case motor so the starting p.f. is very low).

4)



slip ring (phosphor bronze).



→ The brushes are lifted automatically from slip ring and electrically from slip ring. surface. Then the slip ring is shorted out by some gear arrangement automatically.

→ The external resistance is varied under running condition. The speed of IM can be controlled.

→ The slip ring I.M has excellent starting ch. such as high T_{st} less starting current but it has inferior running characteristics such as it has less torque, more exciting current and less no load and full load P.f.

→ Maintenance cost is more. As brush are replaced.

→ Square cage is very rigid construction.

Operation of 3 ϕ I.M:

As the ~~rotor~~ ^{stator} is stationary. So the relative speed between rotor and stator magnetic flux N_s . So according to Farad law of electromagnetic induction ($E_s/p\phi$).

The emf induced in rotor at stand still condition.

$$E_s/p\phi = 4.44 N_p \phi f \text{ kV}$$

When 3 ϕ supply is given to 3 ϕ winding. The stator magnetic field

$$\text{rotates at } N_s \text{ speed. } R = \frac{3\phi}{2} \cos(\theta - \omega t).$$

To satisfy, ω_r 's have the rotor rotates in the same direction as stator rotating magnetic field. But less than N_s speed.
 $\omega_r < \omega_s$. And it is called asynchronous speed.

[If $N = N_s$, Relative speed = 0, $\text{emf} = 0$, $\text{current} = 0$
 $T = 0$]

$N_s - N$ = slip speed of m/c $\Rightarrow s = \frac{N_s - N}{N_s}$

$N = N_s (1-s) \rightarrow$ speed of rotor.

f_2 = freq of rotor at standstill condition.

f_r = " " " running "

$$f_r = s f_2$$

$$f_2 = \frac{N_s P}{120}$$

$$f_r = \frac{(N_s - N) P}{120} \left(\frac{N_s}{N_s} \right) \leftarrow \text{multiply}$$

$$f_r = s f_2$$

Relative speed at running condition.

1) Rotor freq depend upon relative speed:-

full load slip = 2% to 8%. $\Rightarrow f_r = 1$ to 4 Hz at running condition.

$$E_r / \text{ph} = s E_s / \text{ph}$$

$\rightarrow$ Rotor rotating magnetic field (Φ_r) due to emf induced in the rotor winding cause the 120° phase time displacement rotor current and the rotor winding is also 120° space angle. So the rotor rotating magnetic field is produced according to Rotating magnetic field theory.

$\rightarrow$ The resultant rotating magnetic field is also called air gap flux.

$\Rightarrow$ Speed of rotor rotating magnetic field w.r.t. stator = speed of rotor rotating magnetic field w.r.t. rotor + speed of rotor w.r.t. to stator (N) = $N_s - N + N = N_s$.

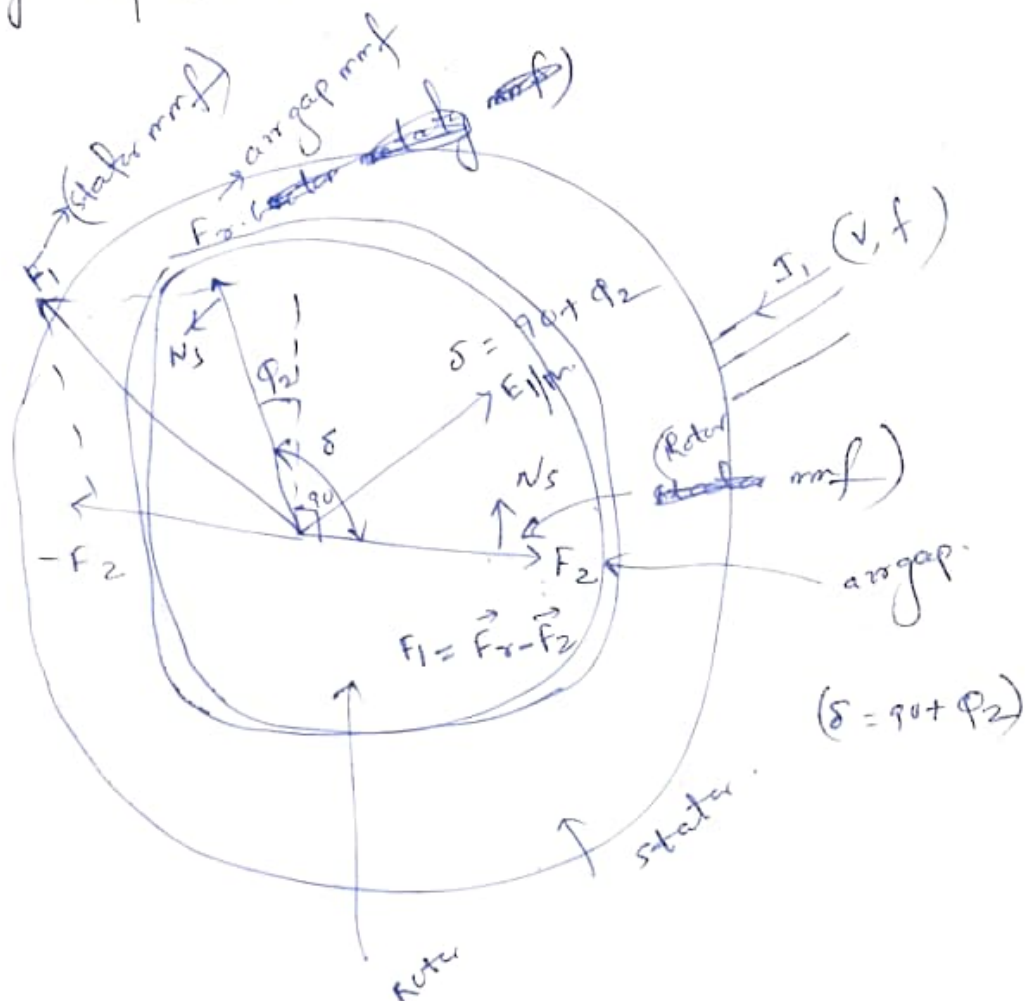
$\rightarrow$ Speed of rotor magnetic field w.r.t. rotor = $\frac{120 f_r}{P}$

$$= \frac{120 s f}{P} = 120 \left(\frac{N_s - N}{N_s} \right) \frac{f}{P} = \frac{N_s (N_s - N)}{N_s} \text{ magnetic field}$$

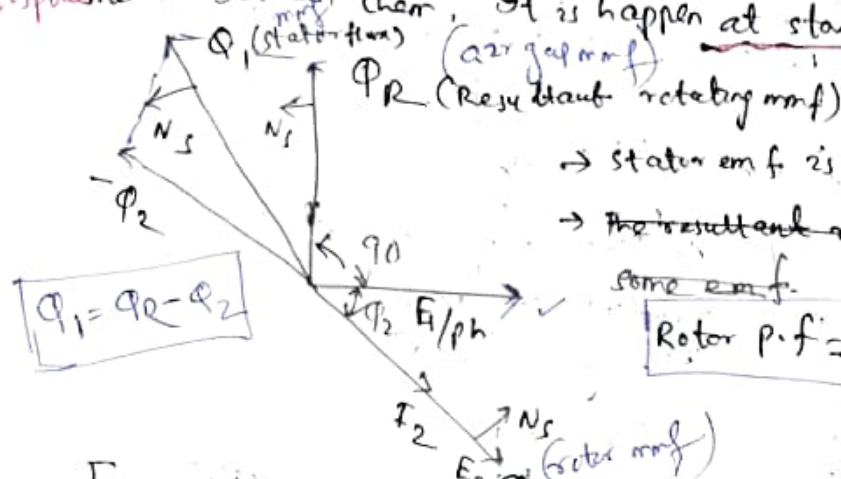
So the relative speed of both rotor and stator is zero.

$\rightarrow$ The stator and rotor rotating magnetic fields rotate at synchronous speed (ω_s) in the air gap and stationary with respect to one another.

for another. That means the relative speed between them is always equal to zero.



→ But some time delay between stator and rotor rotating magnetic field and also the resultant rotating magnetic field. This time delay is called Relative space displacement between them. It happens at starting.



→ stator emf is lagging Φ_R by 90° .

→ The resultant rotating m.f. induces some emf.

$$\text{Rotor P.f.} = R_2 / Z_2 = \cos \phi_2$$

[In T.F. secondary winding P.f. = load P.f.]

But in I.M. the P.f. of rotor depends upon the parameters of rotor. [Rotor resistance is more than rotor reactance] I_2 replaces E_2 .

→ Even though the stator and rotor rotating magnetic fields rotate at synchronous speeds and also stationary with respect to each other (one and other in the airgap). But there is a relative space displacement of $(\phi_2 + 90^\circ)$ between these two magnetic fields.

Because of Relative space displacement cause the torque.

$$T \propto \Phi_R \Phi_2 \sin (90^\circ + \phi_2) \text{ in I.M.}$$

$$T \propto \Phi_1 \Phi_2 \cos \phi_2 \text{ Rotor P.f.}$$

$$T \propto \Phi_1 \Phi_2 \sin \delta \rightarrow \text{Synch motor}$$

Ex. Cage I.M. No rotor winding, only copper bars are inserted.

So the emf induced in rotor (E_2) can't be calculated as there is no winding (only bars). At stand still due to resultant flux (Φ_R) produced in rotor. And rotor current I_2 is found by Lenz's law.

→ stator magnetic field rotates at N_s speed. and produces emf E_1/ph at stand still condition.

→ The production of rotating magnetic field is not according to rotating magnetic field theory. But some other reaction.

is called current sheet principle.

→ The current sheet is represented current distribution at different conductor.

→ change in current distribution due to rotating static magnetic field. So the current distribution changes continuously $(N_s - N)$ speed. These current produced flux, this flux distribution also changes $(N_s - N)$ speed is called rotor rotating magnetic field, $(N_s - N) + N = N_s$.

→ The rotor emf induced at standstill condition.

$$E_2/\text{ph} = 4.44 N_{ph} \Phi_R f \text{ kwr.}$$

$$E_r/\text{ph} = 4.44 N_{ph} \Phi_R s f \text{ kwr.}$$

$$E_s/\text{ph} = s E_2/\text{ph} \rightarrow \text{very small emf induced at running condition.}$$

R_2/ph :- Rotor resistance/ph.

X_2/ph :- Rotor reactance/ph at standstill condition.

$$X_2/\text{ph} = 2\pi f L, \quad X_r/\text{ph} = s X_2/\text{ph}, \text{ under running condition.}$$

$$Z_2/\text{ph} = R_2 + jX_2 = \text{Rotor impedance at standstill condition}$$

$$Z_r/\text{ph} = R_2 + jX_r = R_2 + j(X_2 s) = \sqrt{R_2^2 + (sX_2)^2}$$

$$I_2/\text{ph} = \frac{E_2/\text{ph}}{Z_2/\text{ph}} = \frac{E_2/\text{ph}}{\sqrt{R_2^2 + X_2^2}} \rightarrow \text{at standstill condition.}$$

$$I_r/\text{ph} = \frac{E_r/\text{ph}}{Z_r/\text{ph}} = \frac{s E_2/\text{ph}}{\sqrt{R_2^2 + (sX_2)^2}}$$